

TMHCC DXLibrary Claim Form

Comprehensive Component Documentation

Component	TMHCC_DXLibrary_ClaimForm
Library / Org	DXLibrary / TMHCC
Type	Template (PAGE + CASE)
Stack	React 18 · Cosmos React Core 8.21.3
Pega Infinity	25.1.2-397
Entry point	index.tsx → withConfiguration(...)
Doc date	30 Jul 2026

This document is the full reference for understanding, maintaining, and extending the Claim Form. It reflects the **current source** after recent work including ODE payload handling, drag-and-drop upload, discard exit ownership, case-ID success gate, dialing-code removal, hero blob URL revoke, date display format (`dd/MM/yyyy`), and Add item / Add another item labelling.

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1. Purpose

`TmhccDxLibraryClaimForm` is a five-step wizard that lets a claimant submit a **military kit insurance claim** inside the Pega Constellation portal. It:

- Collects personal, policy, incident, item, and evidence data
- Validates with country-aware and relationship-aware rules
- Uploads supporting evidence to Pega
- Creates a Pega case via a configurable REST path
- Links uploaded attachments to the new case
- Shows a confirmation screen with a Pega-authored note
- Hides surrounding portal chrome so the form reads as standalone

It is a DX **Template** (`type: Template`, `subtype: PAGE + CASE`), so App Studio can place it as a full page and bind case creation.

2. Architecture overview

UI layer

- `index.tsx` — orchestrator (state, wizard, submit, cancel, hero image)
- `sections/*` — one presentational component per step / modal / confirmation

Utils layer

- Validation, dates, file checks, case create, attachment upload/link, localization, portal mask, URL safety

Pega runtime

- `PCore.getRestClient()` , data pages, localization service, asset loader, Cosmos modal manager

Design principles

Principle	How it appears
Single orchestrator	All wizard state in <code>index.tsx</code> (<code>useState</code> / <code>useRef</code>)
Section per step	Presentational sections; parent owns flow
Content-driven rules	Limits, regexes, data-page names live in <code>localizations.json</code>
Defensive Pega use	Fallbacks for missing localization / assets / partial API success
Portal isolation	CSS + <code>MutationObserver</code> hide Constellation chrome

3. File map

```

TMHCC_DXLibrary_ClaimForm/
  index.tsx           Main orchestrator
  types.ts            Shared TypeScript types
  config.json         App Studio descriptor
  localizations.json  UI strings + runtime config values
  countries.json      Bundled country list (ISOCode, Name)
  styles.ts           Global + scoped styled wrapper
  styles/baseStyles.ts Scoped form CSS
  doc-style.css       Print/PDF stylesheet for this document
  COMPONENT_DOCUMENTATION.md This document
  OneColumnPage.svg   Component icon
  mock.ts / mockListValues.ts Storybook mocks
  demo.stories.tsx    Storybook story
  demo.test.tsx       Smoke test (selectors currently stale)
  README.md           Older overview (may lag source)

sections/
  YourDetailsSection.tsx
  WhatHappenedSection.tsx
  ItemsAffectedSection.tsx
  SupportingEvidenceSection.tsx
  ReviewSection.tsx
  SpouseDetailsSection.tsx
  DependentDetailsSection.tsx
  AuthorisedPersonDetailsSection.tsx
  OtherRelationshipDetailsSection.tsx
  PersonContactFields.tsx
  DateTextInput.tsx
  FieldInfo.tsx / InfoIcon.tsx
  DiscardChangesModal.tsx
  ClaimConfirmationView.tsx

utils/
  claimFormValidation.ts
  claimUtils.ts
  dateUtils.ts
  listValuesUtils.ts
  attachmentUploadUtils.ts
  caseAttachmentUtils.ts
  attachmentSessionUtils.ts
  fileValidationUtils.ts
  useLocalization.ts
  usePortalMask.ts
  portalElementUtils.ts
  claimFormStateUtils.ts
  urlUtils.ts
  pegaErrorUtils.ts

```

4. App Studio configuration

From `config.json`:

Key	Value
name / componentKey	TMHCC_DXLibrary_ClaimForm
type	Template

Key	Value
subtype	PAGE, CASE
organization	TMHCC
library	DXLibrary
version	0.0.1
Properties	caseTypeID (TEXT), createClaimApiPath (TEXT), visibility group, About labels
defaultConfig.caseTypeID	TMH-MCS-Work-MilitaryKitClaim
defaultConfig.createClaimApiPath	/api/application/v2/cases
infinityVersion	25.1.2-397
packageCosmosVersion	8.21.3

Data-page names are **not** App Studio props; they live in localization (LovDataPageName , CountriesDataPageName , ConfirmationDataPageName).

Orchestrator prop fallbacks match defaultConfig:

```
caseTypeID = 'TMH-MCS-Work-MilitaryKitClaim'
createClaimApiPath = '/api/application/v2/cases'
```

5. Data model

Defined in types.ts.

UploadedAttachment

id, name, size (bytes), docType, optional docTypeLabel

ClaimItem

id, itemCoverType, itemType, datePurchased, itemDescription, purchasePrice, claimedAmount, optional otherItemType

otherItemType is mapped into the API payload, but no UI currently sets it.

ClaimFormData (by step)

Step	Fields
1 — Your Details	firstName, lastName, email, phoneNumber, policyNumber, serviceNumber, schemeOrProduct, relationship, authorisationConfirmed, policy address (policyCountry, policyAddressLine1/2, policyCity, policyPostalCode), plus conditional spouse / dependent / authorised / other-person fields

Step	Fields
2 — What Happened	<code>dateOfLoss</code> , <code>lossType</code> , <code>otherLossType?</code> , <code>theftReported?</code> , <code>crimeReferenceNumber?</code> , <code>lossLocationType</code> , <code>loss address fields</code> , <code>description</code> , <code>deploymentCountry?</code> , <code>deploymentLocationName?</code>
3 — Items	<code>items: ClaimItem[]</code>
4 — Evidence	<code>attachments: UploadedAttachment[]</code>
5 — Review	<code>declarationConsent: boolean</code>

Defaults (`INITIAL_FORM_DATA`)

- `lossLocationType: 'PHA'`
- `policyCountry: 'GB'`
- `authorisationConfirmed: false`, `declarationConsent: false`
- `items: []`, `attachments: []`

ClaimFormErrors

`Partial<Record<keyof ClaimFormData, string>>`. Step 3 may stuff a per-item error array into `items` via `@ts-ignore`.

6. Orchestrator behaviour

`index.tsx` owns all wizard state and side effects.

State

`formData`, `errors`, `submitError`, `isSubmitting`, `isUploading`, `showConfirmationView`, `createdCaseId`,
`currentStep` (1–5), `listValues`, `countryOptions`, `configLoadError`, `attachmentErrors`,
`attachmentUploadError`, `heroImageUrl`

Refs

`stepContentRef`, `isSubmittingRef`, `isMountedRef`, `skipUnloadWarningRef`

Step navigation

- Nav cards show **steps 1–4 only** (`stepDefinitions`); step 5 is Review with no nav card.
- Footer **Continue** → `handleNext` (validates current step; step 4 also requires each attachment's `docType`).
- Footer **Submit** on step 5 → `handleSubmit` (full validation of steps 1–3). Submit stays disabled until `declarationConsent`.
- No main-flow Back button; Review pencil icons call `setCurrentStep(n)`.
- Step change focuses / scrolls `stepContentRef`.

Field-change side effects

Clearing related fields/errors when `lossType`, `policyCountry`, `lossCountry`, `lossLocationType`, or `relationship` changes. Relationship change clears `RELATED_PERSON_FIELDS` and `authorisationConfirmed`.

List / country load

```
fetchAllListValues(1['LovDataPageName']) // default D_LOVList
fetchCountryList(1['CountriesDataPageName']) // default D_CountryList
```

Either `fetch` failed → `configLoadError` banner (`ErrConfigLoadFailed`) on step 1.

Hero image

`1['HeroImageKey']` → asset loader `getSvcImageUrl`. Blob results use `URL.createObjectURL`. Cleanup **revokes** the object URL on unmount or when `HeroImageKey` changes (including race-safe revoke if load finishes after cleanup).

Submit (`handleSubmit`)

1. Guard with `isSubmittingRef`
2. `getClaimFormValidationErrors` — on failure, set errors and **jump to step 1**
3. `createClaimCase` — require `success` and valid `fullCaseId`
4. `attachUploadedFilesToCase` — on attach failure, set warning (`ErrAttachmentLinkFailedOnConfirmation`) but **still show confirmation**
5. Derive display ID via `getCaseDisplayId` (last token after space), reset form, `clearAttachmentIdsInSession()`

Case ID success gate

`isValidCaseId = /\d+/.test(caseId)`. Invalid / missing ID → `success: false` with `ErrInvalidCaseIdReturned`.

Session attachments

On successful upload/delete: `saveAttachmentIdsInSession(ids)`.

On discard / successful submit: `clearAttachmentIdsInSession()`.

Key: `tmhcc.claimForm.uploadAttachmentIds`. **Write-only today — no hydrate on load.**

7. Five-step journey

Step	Section	Summary
1	<code>YourDetailsSection</code>	Contact, relationship (+ conditional person sections + auth checkbox), policy / service / product, policy address. Info toggle.
2	<code>WhatHappenedSection</code>	Date of loss, incident type (+ OTH / TH extras), description + char counter, location option cards (PHA / ODE / DL) with conditional deployment or loss address.
3	<code>ItemsAffectedSection</code>	Add/edit/save items; saved list; claim summary; max items. Continue disabled when <code>items.length === 0</code> . Button label: Add item when empty, Add another item after ≥1 saved item.

Step	Section	Summary
4	SupportingEvidenceSection	Optional uploads (can submit without evidence); drag-and-drop; per-file evidence type required to Continue if files exist.
5	ReviewSection	Expandable review blocks + Edit jump; declaration checkbox; Submit.

Relationship codes: `PH` (policyholder — no extra section), `SP`, `DP`, `AP`, `OTH`.

Loss location codes: `PHA`, `ODE`, `DL`. Theft loss type: `TH`; other: `OTH`.

8. Section components

Component	Behaviour
Spouse (<code>SP</code>)	Name required; email/phone optional
Dependent (<code>DP</code>)	Name + DOB + free-text relationship required; contact optional
Authorised (<code>AP</code>)	Name + email + phone all required
Other (<code>OTH</code>)	Relationship description + name + contact required; titled "Additional Details"
Authorisation	Checkbox for <code>SP DP AP OTH</code>
DateTextInput	Native <code>type="date"</code> picker only; blocks keyboard/paste/drop; stores localization display format via <code>formatDateForDisplay</code>
FieldInfo	<code>span role="button"</code> (not <code><button></code>) so it does not steal <code><label></code> association
DiscardChangesModal	Cosmos <code>Modal</code> ; <code>confirm</code> → <code>dismiss()</code> then <code>onDiscard()</code> only — does not navigate
ClaimConfirmationView	Loads <code>getPageDataAsync(ConfirmationDataPageName, { pyID })</code> → <code>pyConfirmationNote</code> HTML; optional save <code>.txt</code> reference; Return Home via <code>https://ReturnHomePageLink</code> or parent <code>onClose</code>
PersonContactFields	Digits-only phone capped at <code>PhoneLength</code>
ItemsAffected	Local draft validation via <code>validateItems</code> ; "Item saved" toast (~5s); total claim exceed = UI warning only
SupportingEvidence	Drag/drop + file input; <code>accept={AllowedFileExtensions}</code> ; <code>max</code> <code>MaxAttachmentsCount</code>
Review	Related-person + location-specific fields; declaration + privacy link (<code>PrivacyPolicyLinkUrl</code>)

9. Validation

Central module: `utils/claimFormValidation.ts`.

STEP_FIELDS

- 1: identity, policy, address, all related-person fields, `authorisationConfirmed`
- 2: `dateOfLoss`, `lossType`, `description` in the map; extras via `getWhatHappenedStepErrors`
- 3: `getItemsStepErrors` (not `STEP_FIELDS`)
- 4: no flat validators (doc-type enforced in `handleNext`)

Notable rules

Area	Rule
Names	Min/max from <code>NameMinLength</code> / <code>NameMaxLength</code>
Email	<code>EmailPattern</code>
Phone	Always required; UK format <code>^0\d{PhoneLength-1}\$</code> only if <code>policyCountry === 'GB'</code>
Policy / service numbers	Optional ; format-checked when filled
Dates	<code>DateFormat</code> (default <code>dd/MM/yyyy</code>); not future; purchase $\leq$ date of loss
Amounts	<code>PLAIN_DECIMAL_RX = /^d{1,13}(\.d{1,2})?\$/</code> ; > 0 ; claimed $\leq$ that item's purchase
Postal	Required for GB only; country-specific <code>*_PCPattern</code> / <code>*_PCErrMsg</code>
Theft	<code>lossType === 'TH' → theftReported</code> ; if 'Yes' → <code>crimeReferenceNumber</code>
Location	ODE → <code>deploymentCountry</code> required; DL → full loss address required
Description / item description	Min/max from localization

Total claim warning vs blocking

- **Per-item** claimed $>$ purchase → **blocks** save / step validation
- **Sum** claimed $>$ sum purchase → `ErrTotalClaimExceeds` warning in Items summary — **does not block Continue** (Continue only needs `items.length > 0`)

Full-form submit validation covers steps **1–3 only** (`getClaimFormValidationErrors`).

10. Attachments pipeline

Layer	Details
UI	Drag/drop + file input; max <code>MaxAttachmentsCount</code> (20); disabled at cap
<code>validateFileTypes</code>	Extension or MIME against allow-list; magic bytes for PDF/JPEG/PNG (gif/webp/doc/docx helpers exist but are outside the default allow-list). No client size check despite <code>FileLimitInstruction: "10MB per file"</code>

Layer	Details
<code>uploadAttachments</code>	POST <code>/api/application/v2/attachments/upload</code> (<code>FormData: content, name</code>); concurrency 3 ; 2 attempts; IDs from <code>ID id attachmentID attachmentId</code>
Session	Persist ID array; clear on discard/success
<code>attachUploadedFilesToCase</code>	POST <code>/api/application/v2/cases/\${caseId}/attachments</code> with <code>{ type:'File', category: docType, ID, name }</code>

Evidence type `"Other"`: when `docType === 'Other'`, keeps existing `docTypeLabel` rather than LOV label (can retain a prior label).

11. Case creation payload

```
utils/claimUtils.ts → PCore.getRestClient().invokeCustomRestApi(apiPath, { method:'POST', body }).
```

Top-level content

```
CreationSource: "WebForm", PolicyNumber, DeclarationConsent, Product, ServiceNumber,
RelationshipWithPolicyHolder, OtherRelationship, TotalPurchasePrice (sum of item purchase prices),
UserDetails, PolicyHolder, ClaimInfo, ItemsAffected[]
```

PolicyHolder VS UserDetails

- Always send policyholder contact + policy address as `PolicyHolder`.
- If relationship is `SP / DP / AP / OTH`, `UserDetails` is that person's details + blank `PolicyAddress`.
- `DP` adds `DateOfBirth`; `DP / OTH` set `OtherRelationship`.
- Always adds `AuthorisationConfirmation: String(boolean)`.

ODE `LossAddress` special case

When `lossLocationType === 'ODE'`:

```
{ BaseLocation: deploymentLocationName || '', pyCountry: deploymentCountry || '' }
```

- `PHA` copies policy address fields into the standard loss-address shape.
- `DL` (and non-ODE non-PHA) uses loss address fields (`pyCity, pyCountry, pyPostalCode, pyStreetAddress, pyStreetAddress2`).

Dates → ISO

`toIsoDate(value, getDateFormat(1))` for `LossDate, item PurchaseDate`, dependent `DOB`; falls back to raw string if conversion fails.

Items map includes `OtherItemType: item.otherItemType` even though no UI sets it today.

12. Pega integrations

API	Usage
<code>PCore.getDataPageUtils().getDataAsync</code>	<code>D_LOVList (select pyKeyString, pyActualValue, pyCategory, pyRowNumber, SupportingText);</code> <code>D_CountryList (Name, ISOCode only — no dialing code)</code>
<code>PCore.getDataPageUtils().getPageDataAsync</code>	Confirmation note
<code>PCore.getRestClient().invokeCustomRestApi</code>	Create case, upload, attach
<code>PCore.getAssetLoader().getSvcImageUrl</code> (with <code>AssetLoader</code> fallbacks)	Hero image
<code>getPConnect().getLocalizationService().getLocalizedText</code>	Override <code>localizations.json</code>
Cosmos	Banner, Icon, Progress, Modal, useModalManager, withConfiguration

LOV categories (`LIST_CATEGORIES`): `Product`, `Relationship`, `LossType`, `LossLocation`, `GenericYesNo`, `ItemType`, `ItemCoverType`, `EvidenceType`.

`CountryOption` is a `ListOption` alias (`key / label / optional subtitle`). Dialing-code support has been removed from types, fetch, mock data, and styles.

13. Localization and config values

Notable keys in `localizations.json` (defaults):

Key	Default
<code>DateFormat / DateFormatPlaceholder</code>	<code>dd/MM/yyyy</code>
<code>ErrDateFormatInvalid</code>	Enter a valid date in {1} format
<code>HeroImageKey</code>	<code>WEBWB!HOMEPAGEIMAGE!PNG</code>
<code>LovDataPageName</code>	<code>D_LOVList</code>
<code>CountriesDataPageName</code>	<code>D_CountryList</code>
<code>ConfirmationDataPageName</code>	<code>D_ConfirmationMessage</code>
<code>MaxAttachmentsCount</code>	20
<code>MaxItemsCount</code>	20
<code>AllowedFileExtensions</code>	<code>.jpg, .jpeg, .pdf, .png</code>
<code>AllowedFileTypesLabel</code>	JPG, PNG, PDF
<code>FileLimitInstruction</code>	10MB per file (display only)
<code>EmailPattern / PolicyNumberPattern / ServiceNumberPattern</code>	see JSON

Key	Default
<code>PostalCodePattern</code> / <code>GB_PCPattern</code> / <code>US_PCPattern</code>	country-aware
<code>PostalCodeStripPattern</code>	<code>[^a-zA-Z0-9]</code>
Lengths	<code>NameMin/MaxLength</code> 2/100, <code>Description</code> 10/500, <code>ItemDescription</code> 5/500, <code>PhoneLength</code> 11, city limits
<code>CharCounterWarnThreshold</code>	475
<code>Add item</code> / <code>AddAnotherItem</code>	Add item / Add another item
<code>ReturnHomePageLink</code>	<code>https://talktotrinity.com/</code>
<code>PrivacyPolicyLinkUrl</code>	<code>https://talktotrinity.com/security-and-privacy/</code>

Dates in form state are stored in `DateFormat`. Native date inputs bridge via `utils/dateUtils.ts` (`toIsoDate`, `formatDateForDisplay`, `compareDates`, `isFutureDate`, etc.). `applyDateInputMask` exists but is unused by the current picker-only `DateTextInput`.

14. Portal masking and styling

- `usePortalMask(300)` → `startHidingPortalElements()` (CSS injection + `MutationObserver` hiding summary-item header and account menu) + overlay `claim-form__portal-mask` with `Progress` for 300ms.
- Root `#tmhcc-claim-form-root`; global `*:has(#tmhcc-claim-form-root) { padding: 0 !important }` to defeat Pega harness padding.
- Two-column page: hero image panel + form panel; discard modal backdrop scoped via `:has(.claim-form__discard-modal-box)`.
- Font: `var(--pega-font-family, 'Open Sans', Arial, sans-serif)`.
- Phone UI is a single input inside `.claim-form__phone-group` (dialing-code select styles removed).

15. Accessibility and security

Accessibility

- Step cards: `aria-current="step"`
- Errors: `role="alert"`, often `aria-invalid` + `aria-describedby`
- Required: `aria-required`, visual *
- Info toggles: `aria-expanded` / `aria-controls` / show-hide labels
- Review headers keyboard-activatable (`Enter` / `Space`)
- Delete attachment: `aria-label` with filename
- Portal mask: `aria-hidden="true"`
- Form panel: `tabIndex=-1` for programmatic focus on step change
- `FieldInfo` intentionally not a native button inside labels

Security / safety

- `isSafeUrl` — https only for Return Home / privacy links
- File type validation includes magic-byte checks for allowed types
- Unload warning when form is dirty (skipped when discard intentionally exits via `skipUnloadWarningRef`)

16. Cancel, discard, confirmation

Action	Who navigates
Cancel → “No, Continue”	Modal <code>dismiss()</code> only
Cancel → “Yes, Cancel Form”	Parent <code>handleDiscardAndClose</code> → <code>closeCurrentBrowserContext(https ReturnHome...)</code>
Confirmation “Return to Home Page”	<code>ClaimConfirmationView</code> → <code>window.location.href</code> if safe URL, else parent <code>onClose</code>
Confirmation after attach warning	Still confirmation; warning banner only

Discard navigation is owned **solely by the parent** to avoid double-exit (modal previously navigating as well).

17. Storybook and tests

Storybook (`demo.stories.tsx`)

Stubs REST (upload IDs, case `CLM-1234`), data pages via `mockGetDataAsync` / `mockListValues` , asset loader URL, localization from JSON + `HeroImageKey` override.

Tests (`demo.test.tsx`) — stale

Current assertions expect:

- `'Claim Form'`, `'Your details'`, `'Supporting Evidence'`, button `'Submit Claim'`

Live UI uses localized `'Your Details'`, `'Evidence'`, and footer `continue` on step 1 (Submit only on review). Smoke test will fail against current strings/flow until refreshed.

18. Design decisions (intentional)

- Policy / service numbers optional but format-validated when present
- Non-GB phones: presence only
- Spouse / Dependent contact optional; Authorised / Other required
- Clear related-person state when relationship changes
- Total claim vs purchase overage is warning-only; per-item overage blocks
- Evidence optional to submit, but if files exist each needs a type before leaving step 4
- Attach failure after case create still shows confirmation + warning

- Discard navigation owned solely by parent
- `skipUnloadWarningRef` set synchronously before unload/close
- `FieldInfo` uses `role="button"` span to preserve label → input association
- Date entry via native picker only
- Upload concurrency capped; case-attach has no retry
- Country options have **no dialing code** (phone dialing UI removed)
- Add-item button label switches after the first saved item
- Hero blob URLs are revoked on unmount / key change

19. Recommended further enhancements

Prioritized backlog (not blockers for the current production path):

High

1. **Refresh** `demo.test.tsx` **selectors** to match live strings (`Your Details` , `Evidence` , `Continue / Submit`) and restore a useful smoke test.
2. **Client-side 10MB file size check** — UI copy says 10MB per file, but `fileValidationUtils` currently checks type/magic bytes only.

Medium

1. **Submit validation jump** — on full-form errors, jump to the first failing step (not always step 1).
2. **Hydrate session attachment IDs** after refresh, or stop writing IDs until hydrate exists (avoid orphaned session entries).
3. **otherItemType UI** — field + API mapping exist; no input in `ItemsAffectedSection` / `createBlankItem` (deferred by design for now).

Low

1. **Evidence type “Other”** — avoid retaining a stale prior `docTypeLabel` in Review.
2. **otherLossType blur validator** — currently validated mainly in step-level logic.
3. **Unused App Studio title prop** — wire it or remove from config properties.
4. **applyDateInputMask** — unused after picker-only date inputs; remove or reintroduce typed entry.
5. **README lag** — older README still mentions dialing codes in places; prefer this document as source of truth.

20. How to extend

Add a form field

1. Add to `ClaimFormData` / `INITIAL_FORM_DATA`
2. Add localization label / errors
3. Register validator (+ `STEP_FIELDS` / step helper if step-gated)
4. Render in section
5. Map into `createClaimCase` if needed

6. Show on Review if users must verify it

Add an LOV dropdown

1. Ensure category exists on `D_LOVList`
2. Wire `listValues[LIST_CATEGORIES....]`
3. Validate and include in payload

Add an App Studio property

1. Add to `config.json` `properties` / `defaultConfig`
2. Destructure in `index.tsx` with a safe default

21. Glossary

Term	Meaning
PHA	Policyholder address — loss happened at usual address
ODE	On deployment / exercise — country + optional base/location only in payload
DL	Different location — full loss address required
SP / DP / AP / OTH	Spouse / Dependent / Authorised Person / Other relationship
TH	Theft loss type
Constellation	Pega DX / Cosmos portal runtime
LOV	List of values from a Pega data page
Object URL	Browser blob URL from <code>URL.createObjectURL</code> ; must be revoked when done

End of document.